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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **MLA02**

Course Name: **FUNDAMENTALS OF MACHINE LEARNING**

Duration: **1:30 Hr**

Date : 31/10/2026

Max Marks: **50 Marks**

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

X \ Y	Y = 0	Y = 1
X = 2	0.20	0.10

Analyse the distribution by (a) finding the marginals of X and Y, (b) computing E[X], Var(X) and E[Y], (c) computing $P(X = 2 \mid Y = 1)$, (d) computing Cov(X, Y) and the correlation coefficient, and (e) deciding whether X and Y are independent, with a justification.

(6) (CO1) (BL4)

Task 4. A temperature sensor output is modelled as $X \sim N(\mu = 50, \sigma^2 = 16)$.

(a) Calculate $P(X > 58)$ and $P(46 < X < 54)$.

(b) A second sensor gives the readings 12, 15, 11, 14, 13. Derive the maximum-likelihood estimates of its mean and variance and compare the ML variance with the unbiased estimate, stating which is biased and why. (Use $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.)

(4) (CO1) (BL4)

Task 5. A source emits four symbols with probabilities (1/2, 1/4, 1/8, 1/8). Calculate its entropy, compare it with the entropy of a uniform source over four symbols, and analyse the redundancy. Also compute the average length of the code {0, 10, 110, 111} and comment on its optimality.

(3) (CO1) (BL4)

Task 6. A model predicts class probabilities Q = (1/3, 1/3, 1/3) while the true distribution is P = (0.5, 0.25, 0.25). Compute KL(P‖Q) and KL(Q‖P) in bits, and analyse what the asymmetry implies when KL divergence is used as a training loss.

(3) (CO1) (BL4)

Task 7. Two binary variables A and B have the joint distribution $P(0,0) = 0.4$, $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$. Calculate H(A), H(B), the joint entropy H(A,B), the conditional entropy H(A|B) and the mutual information I(A;B). Analyse what the value of I(A;B) says about using B as a feature to predict A.

(4) (CO1) (BL4)

Task 8. A one-dimensional feature x follows $p(x|\omega_1) = N(2, 4)$ for class ω_1 and $p(x|\omega_2) = N(6, 4)$ for class ω_2, with priors $P(\omega_1) = 0.6$ and $P(\omega_2) = 0.4$ and zero–one loss. Derive the Bayes decision threshold x^*, classify x = 4.5, and calculate the overall probability of error. Analyse how the threshold moves as the prior of ω_1 increases.

(4) (CO1) (BL4)

Task 9. For an observation x, the posteriors are $P(\omega_1|x) = 0.3$ (disease) and $P(\omega_2|x) = 0.7$ (healthy). Action α_1 = 'declare disease', α_2 = 'declare healthy'. The loss matrix is $\lambda(\alpha_1|\omega_1) = 0$, $\lambda(\alpha_1|\omega_2) = 8$ (unnecessary treatment), $\lambda(\alpha_2|\omega_1) = 50$ (missed disease), $\lambda(\alpha_2|\omega_2) = 0$. Evaluate the conditional risks, decide the minimum-risk action, compare it with the MAP decision and determine the posterior $P(\omega_1|x)$ above which α_1 is optimal.

(3) (CO1) (BL5)

Task 10. Polynomial regression models of increasing degree were fitted to the same dataset, giving the mean-squared errors below.

Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1

Degree	1	2	3	5	9
Validation MSE	9.5	4.6	2.9	3.6	11.8

Calculate the generalisation gap for each degree, diagnose under-fitting/over-fitting using the bias–variance trade-off, select the best model with justification, and recommend two techniques to improve the degree-9 model.

(4) (CO1) (BL5)

Task 11. A coin is tossed 10 times and shows 7 heads. The bias θ = P(head) is given a Beta(2, 2) prior. Calculate the ML estimate, the MAP estimate and the posterior mean of θ. Then evaluate how the three estimates change if the data were 70 heads in 100 tosses, and explain the role of the prior.

(4) (CO1) (BL5)

Task 12. A hospital wants an early-warning system that raises an alert when an ICU patient is likely to develop sepsis within the next 6 hours. Historical electronic health records (vital signs, lab values, outcomes) are available.

(a) Design the learning system: state the task T, performance measure P and experience E, the target function, a suitable input representation, a learning algorithm and the validation strategy. [3]

(b) On a validation cohort of 1,000 patients, 5% develop sepsis. The trained model has sensitivity 0.90 and specificity 0.80. Compute TP, FN, FP and TN. [1]

(c) Compute the positive predictive value (PPV) and negative predictive value (NPV). [1]

(d) A missed sepsis case costs ₹100 (unit cost) and a false alarm costs ₹5. Compare the total cost of (i) using the model, (ii) alerting every patient and (iii) never alerting. [2]

(e) Recommend a deployment policy and justify how the alert threshold should be set using Bayes decision theory. [1]

(8) (CO1) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - I

Programme	B.E / B.Tech	**Course Code**	**MLA02**
Course Name	**Fundamentals of Machine learning.**	**Date**	3/10/2026
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: B. Maheshwar (192411195)

TASK 1	Max Marks	Marks Awarded
Task 1	5	
Task 2	2	
Task 3	6	
Task 4	4	
Task 5	3	
Task 6	3	
Task 7	4	
Task 8	4	
Task 9	3	
Task 10	4	
Task 11	4	
Task 12	8	
TOTAL	**50**	

Total in words:

46/50

Fundamentals of Machine Learning:

Here is the compressed version of all 12 tasks. I have kept the important formulas, calculation steps and final answers without lengthy explanation

Task-2 · Design of learning systems

Case	Learning type	Task
(a) loan default	supervised	classification
(b) customer segments	unsupervised	clustering
(c) credit-card spend	supervised	Regression
(d) ATM cash refill	Reinforcement	control
(e) fraud detection	semi-supervised	classification

T, E, P:

- (a) T: predict default; E: Labelled records; P: Accuracy
- (b) T: Group customers; E: Unlabelled data; P: clustering quality
- (c) T: predict spending; E: past spending; P: MSE
- (d) T: Learn refill policy; E: Trial and feedback; P: Reward
- (e) T: Detect fraud; E: Labelled and unlabelled data; P: precision and recall.

Label Noise: Loan default Prediction is highly Affected by incorrect labels.

Task 2: Baye's Theorem

given: prevalence = 0.02, sensitivity = 0.95, specifically =

$$P(D|+) = \frac{0.95(0.02)}{0.95(0.02)+0.10(0.98)} = \frac{0.019}{0.117}$$

$$= 16.24\%$$

Conclusion: low disease prevalence causes more false positives.

Task 3. Joint probability distribution

X	Y = 0	Y = 1
0	0.10	0.20
1	0.15	0.25
2	0.20	0.10

(a) margins

$$P(X) = (0.30, 0.40, 0.30)$$

$$P(Y) = (0.45, 0.55)$$

(b) Expectations and variance

$$E(X) = 0(0.3) + 1(0.4) + 2(0.3) = 1$$

$$var(X) = 1.6 - 1^2 = 0.6$$

$$E(Y) = 0(0.45) + 1(0.55) = 0.55$$

(c) Conditional probability

$$P(X=2|Y=1) = \frac{0.10}{0.55} = 0.1818$$

(d) Covariance and correlation

$$Cov(X,Y) = 0.45 - (1)(0.55) = -0.10$$

$$\rho = \frac{-0.10}{\sqrt{0.6(0.2475)}} = -0.2593$$

6

(e) Independence

$$0.10 \neq 0.30(0.45) = 0.135$$

Therefore, X and Y are not independent

Task 4. Normal Distribution and MLE

$\mu = 50, \sigma = 4$

(a) probabilities

1. $$z = \frac{58 - 50}{4} = 2$$

$$P(X > 58) = 1 - 0.9772 = 0.0228$$

2. $$P(46 < X < 54) = 0.8413 - 0.1587 = 0.6826$$

(b) MLE

Readings: 12, 15, 11, 14, 13

$$\bar{x} = \frac{65}{5} = 13$$

14

$$\sigma^2 = \frac{1+4+4+1+0}{5} = 2$$

Unbiased variance:

$$s^2 = \frac{10}{4} = 2.5$$

Conclusion: ML variance is biased because it divided by n, while unbiased variance divide by $n-1$

Task 5. Entropy

given: $P = (1/2, 1/4, 1/8, 1/8)$

a) Entropy:

$$H(x) = -\sum P(x) \log_2 P(x)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$= 1.75 \text{ bits}$$

b) Uniform entropy and redundancy

$$H = \log_2 4 = 2 \text{ bits}$$

$$R = 2 - 1.75 = 0.25 \text{ bits}$$

c) Average code length

Code: 1, 01, 001, 000

$$L = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$$

$$\boxed{L = 1.75 \text{ bits}}$$

Conclusion: code is optimal because average length equal entropy

Task 6. KL Divergence

$P = (0.5, 0.25, 0.25)$.

$q = (1/3, 1/3, 1/3)$

(a)

$$D(P||Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)}$$

$$D(P||Q) = 0.0850 \text{ bits}$$

(b)

$$D(Q||P) = \sum Q(x) \log_2 \frac{Q(x)}{P(x)}$$

$$D(Q||P) = 0.0817 \text{ bits}$$

conclusion:

KL divergence is asymmetric because reversing P and Q changes its value

Task 7: Entropy and Mutual information

$P(0,0) = 0.4$, $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$

(a) Individual entropy

$H(A) = H(B) = 1$ bit

(b) Joint entropy

$H(A,B) = -[2(0.4 \log_2 0.4) + 2[0.1 \log_2 (0.1)]$

1.722 bit.

(c) Conditional entropy

$H(A|B) = 1.722 - 1 = 0.722$

(d) Mutual information

$I(A;B) = 1 - 0.722$

conclusion: B provides useful information for predicting A

Task 8. Bayes Decision Theory

$\mu_1 = 2$, $\mu_2 = 6$, $\sigma^2 = 4$, $P(w_1) = 0.6$

(a) Decision threshold

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{0.6}{0.4}$$

$$= 4 + \ln(1.5)$$

$$x^* = 4.4055$$

(b) classification

$4.5 > 4.4055$

w_2

(c) probability of error

$P(error) = 0.6(0.1146) + 0.4(0.2127)$

0.154

(d) Increasing the prior of w_1 increases the threshold and it decision region

Task 9: Minimum Risk decision

$P(w_1|x) = 0.3$, $P(w_2|x) = 0.7$

a) Risk of declaring disease

$R(\alpha_1) = 0(0.3) + 8(0.7)$

$R(\alpha_1) = 5.6$

b) Risk of declaring healthy

$R(\alpha_2) = 50(0.3) + 0(0.7)$

$R(\alpha_2) = 15$

c) minimum risk

$5.6 < 15$

Declare disease.

d) MAP decision:

MAP declares healthy because 0.7 is greater than 0.3. minimum-risk decision also declares disease

(e) posterior threshold

$8(1-P) < 50P$

$8 < 58P$

$\boxed{P > 0.1379}$

Task 10. Bias - variance Trade-off

Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1
Validation MSE	9.5	4.6	2.9	3.6	17.8
Gap	0.5	0.5	0.7	2.7	11.7

(a)

Gap = Validation MSE - Training MSE

(b) model diagnosis

- Degree 1, 2 : under-fitting
- Degree 3: Good balance
- Degree 5, 9 : over-fitting

(c) Best model

Degree 3

Reason : Lowest validation MSE (2.9).

(d) Improvement techniques

1. Regularisation (L1/L2)
2. Increase training data.

Task 11. Bayesian parameter Estimation

given 7 heads, 3 tails, prior Beta(2, 2)

(a) MLE

$$\hat{\theta} = \frac{7}{10} = 0.7$$

(b) MAP

$$\text{Beta}(1+7, 2+3) = \text{Beta}(9, 5)$$

$$\hat{\theta}_{MAP} = \frac{9-1}{9+5-2}$$

$$\boxed{0.6667}$$

(c) posterior mean

$$E(\theta) = \frac{9}{14} = 0.6429$$

(d) for 70 heads in 100 tosses

Estimate	value
MLE	0.7000
MAP	0.6961
posterior mean	0.6923

conclusion: with more data, the influence of the prior decreases.

Task 12. Hospital early-warning system

(a) Design the learning system

- T: predict sepsis within 6 hours
- P: sensitivity, specificity and F1-score
- E: Historical patient records.

Target : Probability of sepsis

Input : vital signs and laboratory values

Algorithm : logistic regression

validation : 70% - test split and cross-validation.

(b) confusion matrix

given : 1000 patients, pb = above = 5%, sensitivity = 90%,

specificity = 80%

Actual positive : 1000 (0.05) = 50

actual negative :

1000 - 50 = 950

TP : 50(0.9) = 45

FN = 50 - 45 = 5

TN = 950(0.8) = 760

FP = 950 - 760 = 190

	Actual sepsis	Actual healthy
Predicted sepsis	45	190
Predicted healthy	5	760

(c) PPV and NPV

$$PPV = \frac{45}{45 + 190} \times 100$$

$$PPV = 19.15\%$$

$$NPV = \frac{760}{760+5} \times 100$$

$$NPV = 99.35\%$$

(d) Total cost

using model

$$(5 \times 100) + (190 \times 5) = 1450$$

$$950 \times 5 = 4750$$

$$50 \times 100 = 5000$$

using Bayes decision theory

$$5(1-P) < 100P$$

$$5 < 105P$$

$$P > 0.0476 = 4.76\%$$

Conclusion:

Deploy the model with clinical supervision because it has the lowest cost. Set the alert threshold considering the high cost of missed sepsis